3 The blank must first be bored out to give clearance for the ink converter and then to the correct size for the 8.5mm tap. Getting the length correct is vital; too short and the components will not fit together, too long and you risk breaking through when parting off the end of the barrel

4 The diameter of the bore is important too. Enough clearance is required to prevent the ink converter from binding in the barrel, but the smaller the bore, the greater will be the strength of the acrylic body. Remember there is no brass tube to provide additional strength. I measured the converter at various positions with digital calipers and drilled out in stages to maximise the thickness of the body material

5 As always when drilling acrylic materials my advice is slow speeds, sharp drill bits and frequent withdrawal of the bit to clear swarf. When drilling deep holes where the depth of the bore exceeds the length of the drill bit flutes withdrawal is particularly important and is required for every few millimetres of drilling as the swarf will quickly build up and block the flutes. Excessive heat build-up from blocked flutes can cause the blank to melt irretrievably to the drill bit as I know from my early experiences with acrylic materials

6 I set the initial 6.5mm drill bit into the chuck so that full depth would be reached when the chuck jaws just touched the end face of the blank, but for subsequent bits the appropriate depth for each was measured and marked with masking tape. You could measure the distance moved by the tailstock quill for each turn of the handwheel and calibrate the depth in this way if you choose. The final drill bit should be the correct size for the tap that you intend to use. Check with tapping drill charts or against the manufacturer's specification

7 Once drilled to size for the internal thread I initially mounted the tap in a keyless chuck and began to create the threads by turning the headstock handwheel whilst maintaining light pressure on the tailstock with my other hand then reversing the handwheel to clear the swarf and thus gradually cutting the thread little by little. I found this to be a little troublesome as when reversing the tap it tended to come loose so I changed over to a keyed Jacobs chuck which worked much better as would a collet chuck or similar. Remember that you are working with a much more delicate material than the aluminium used for the section so gently does it

8 I blew out all the swarf with an airline and then trial fitted the section and ink converter to make sure that everything was a good fit

9 The exposed part of the blank can now be turned down in preparation for cutting the spigot where the threads will be formed on the

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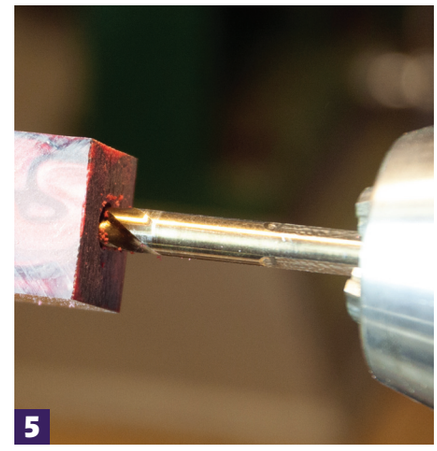
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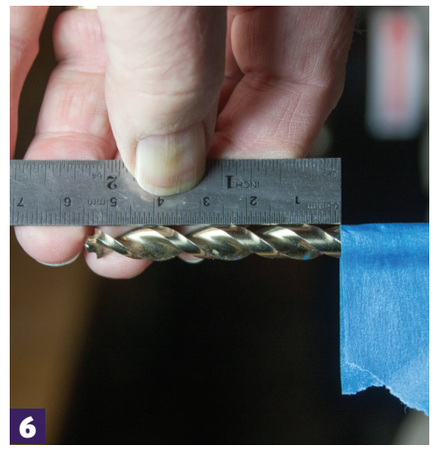
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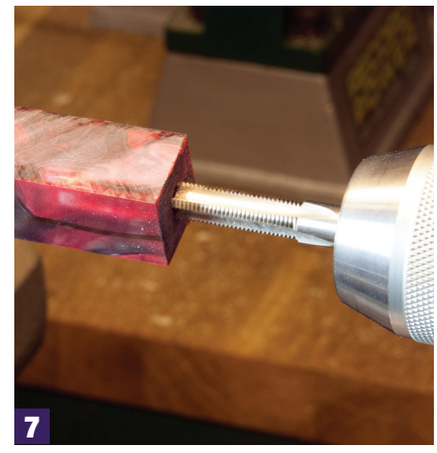
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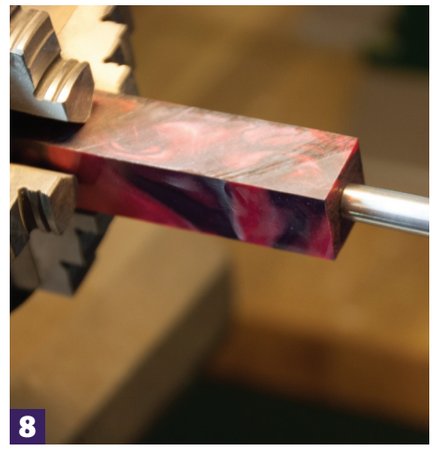
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